Calgary Fuel Price Forecasting

GUIDED STUDENT WORKSHEET | AI-Assisted Time-Series ML | Instructor-Led

Objective: Predict the next recorded business-day Calgary regular gasoline price (cents per litre) using historical prices. Follow your instructor in one AI conversation. Save each file as you go.

Data: Use the full provided Calgary gasoline CSV through October 20, 2026.. Do not assume the file contains a price for October 21.(01/01/2021 - 20/10/2026)

1 Data Collection & Problem

Why? Fuel-cost estimates can help construction and logistics teams plan transportation budgets.

1. Open Kalibrate historical charts (https://charting.kalibrate.com/Charting/FullMonty). Locate Calgary, regular unleaded, retail price and daily observations.
2. Upload the full supplied CSV. Identify the business problem, input and target.

COPY PROMPT Explain the business value of forecasting Calgary gasoline prices. State the inputs, predicted output and one limitation in plain language.

CHECK / RECORD: What decision could this forecast support?

Refuel fleet vehicles strategically and prepare more reliable fuel budgets for construction and logistics operations.

DP1 Data Overview

Understand the dataset before changing anything.

3. Inspect the row count, date range, column names and units.
4. Calculate mean, median, minimum and maximum prices. Plot prices over time and annual averages.
5. Describe two patterns or changes you notice.

COPY PROMPT Overview my uploaded fuel-price dataset. Show its columns, date range, summary statistics, missing values and a price-over-time chart. Explain the patterns simply.

CHECK / RECORD: Row count, date range, price range and two observations.

1,516 rows; Jan 1, 2021 - Oct 23, 2026; 100.2-191.9 cents/L.
2022 had the highest prices; price levels rose again in 2026.

DP2 Data Wrangling

Clean the observations. Let AI investigate and recommend the historical window and lag features.

6. Check date formats, chronological order, missing/duplicate/invalid prices and survey frequency. Do not invent weekends.
7. Ask AI to analyze patterns and recommend a suitable window size and lag features. Window size = how many earlier observations are considered; lag = a specific earlier price (lag_1 is the last recorded price).
8. Ask why the recommended values suit these data, what alternatives were considered and how to avoid using future prices.
9. Create the recommended lag columns, inspect five rows and download cleaned_data.csv and lag_features.csv.

COPY PROMPT Clean my fuel-price dataset without inventing weekend data. Analyze historical patterns and recommend a window size and lag features for next-recorded-day prediction. Explain the reasons and alternatives using past data only. Create the lag features and give me cleaned_data.csv and lag_features.csv to download.

CHECK / RECORD: Recommended window: _____; lags: _____; reasons: _____.

Window: 5 recorded days; lags: 1, 2, 3, 5.
Reason: strong short-term persistence; lag 1 had lowest simple-rule error.
Five observations capture recent direction and volatility without weekend filling.

DP3 Data Transformation

Create modeling inputs and a fair time-ordered train/validation split.

10. Set X = AI-selected lag features and y = current Unleaded price. Retain Dates for checking and charts.
11. Split chronologically (earlier observations for training, later observations for validation); do not randomly shuffle.
12. Display dates and row counts for both groups; download model_train.csv and model_validation.csv.

COPY PROMPT Using my selected lag features, perform standardization or normalization. define X and the target price y. Split the dataset into earlier training rows and later validation rows without shuffling. Explain the split and give me both CSV files.

CHECK / RECORD: Training date range: _____; validation date range: _____. Why must time order be preserved?

Training: Jan 8, 2021 - Aug 26, 2025 (1,208 rows).
Validation: Aug 27, 2025 - Oct 23, 2026 (303 rows).
Keep time order to simulate future prediction and prevent leakage.

5 Modeling

Use the same input features and training rows to compare four regression approaches.

13. Try Linear Regression (simple/interpretable), Decision Tree (rules), Random Forest (multiple trees) and Gradient Boosting (sequential trees). Other regression algorithms are allowed with reasons.
14. Add the baseline: predict the next price as equal to the previous recorded price.
15. Train on training rows only. Save the fitted models and note the algorithms/settings.

COPY PROMPT Train Linear Regression, Decision Tree, Random Forest and Gradient Boosting on my training file using the chosen lag features. You may recommend alternatives with reasons. Add a previous-price baseline. Save the trained models.

CHECK / RECORD: Why were these algorithms chosen?

Linear Regression: clear benchmark; Decision Tree: nonlinear rules;
Random Forest: stable ensemble; Gradient Boosting: sequential corrections.
Ridge tests correlated lags; previous-price forecast sets the baseline.

6 Validation

Compare models on later historical records, using only information available before each predicted date.

16. Predict validation prices and compare with actual prices.
17. Calculate MAE (main metric), RMSE and RMSLE for each model and the previous-price baseline.
18. Create an actual-versus-predicted chart. Select the model with the lowest validation MAE; download validation_predictions.csv and model_comparison.csv.

COPY PROMPT Use the validation file to compare actual and predicted prices for every model and the previous-price baseline. Calculate MAE, RMSE MedAE RMSLE, plot actual vs predicted, and recommend the model with the lowest MAE. Download the results as CSV files.

CHECK / RECORD: Best model: _____; MAE: _____ cents/L; baseline MAE: _____.

Best model: Linear Regression; MAE: 3.320 cents/L;
previous-price baseline MAE: 3.653 cents/L (303 validation rows).

7 AI Governance

Check whether the proposed system can be used responsibly.

19. Review data quality, leakage, changing market prices, licensing and human oversight.
20. Describe how to monitor forecasting error over time.

COPY PROMPT Create a short governance checklist for this fuel-price model covering data quality, leakage, licensing, market changes, monitoring and human review.

CHECK / RECORD: One key risk and how to manage it.

Risk: sudden refinery outages, oil shocks or tax changes may break
historical patterns. Track error vs baseline; require human approval.

8 Deployment — Simple HTML Website + Working Predictor

The final output should be an easy-to-open single HTML page, not a complicated application.

21. After selecting the best model, refit it on all available historical records through October 19, 2026 using the same selected lag features.
22. Build a simple HTML page that shows the complete workflow: data collection, DP1, DP2, DP3, Modeling, Validation, Governance and Deployment, with brief descriptions, charts and results.
23. Include a working forecasting section: users enter the required number of earlier prices AND their dates; show which dates/values feed each chosen lag; click Predict to calculate the next recorded-business-day gasoline price in cents/L using the REAL fitted model or its exported parameters.
24. The prediction must change when inputs change. Validate dates and numeric prices. Do not use a fixed sample answer or a random forecast.
25. Add working download CSV buttons for cleaned data, lag features, train/validation files, validation predictions and model comparison. Embed files in the HTML or package all files together. Open each download to verify it works.

26. Open index.html in a browser, test two sets of inputs, verify the predictions and save a complete website package.

    COPY PROMPT Create a simple standalone HTML website showing our complete Calgary fuel-price ML workflow (Data Collection, DP1–DP3, Modeling, Validation, AI Governance and Deployment), with brief explanations, charts and actual results. Include a working forecasting app where users enter dated past prices and click Predict to get the next recorded-day price using our final trained model and selected lags. Add working downloads for all generated CSV files. Test the predictions and downloads, then provide the HTML website.

CHECK / RECORD: HTML opens: yes/no; prediction changes when input changes: yes/no; CSV downloads work: yes/no.

HTML opens: YES (local browser test); inputs alter forecast: YES;
CSV downloads: YES (embedded tests). Public hosting: NOT done.
Note: worksheet cuts off Oct 20; our analysis extends to Oct 23, 2026.

Submission checklist

☑ Completed explanations and recorded observations

☑ Cleaned, lagged, train and validation CSVs

☑ Four-model (or justified alternative) comparison with MAE and actual-versus-predicted chart

☑ Governance checklist

☑ Simple HTML site showing all steps AND a working dated-price prediction interface; verified downloads

Evidence: model files, validation comparison, HTML and CSVs generated.
Caution: independent unseen testing and prescribed Oct 19 refit not verified.
Public web deployment also remains pending.

Instructor note: This is a guided demonstration. Confirm AI-generated results with the actual files. After final validation and model selection, fit the deployment model using all available records. The absence of a separate unseen test limits evidence of future performance.